IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc Communications and Signal Processing

DIGITAL IMAGE PROCESSING

Wednesday, 8 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. (a) Let $f(x,y)$ denote an $N \times N -$ point two-dimensional sequence, that has zero value outside $0 \le x \le N-1,\ 0 \le y \le N-1$, where N is an integer power of 2. In implementing the standard Walsh Transform of $f(x,y)$, we relate $f(x,y)$ to a new $N \times N -$ point sequence $W(u,v)$.

(i) Define the sequence $W(u,v)$ in terms of $f(x,y)$. Explain all the symbols used. **[1]**

(ii) For the case $N = 2$ and $f(x,y) = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ calculate the Walsh transform coefficients. **[4]**

(b) Consider the population of vectors $\underline{f}$ of the form

$$\underline{f}(x,y) = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ \vdots \\ f_N(x,y) \end{bmatrix}.$$

Each component $f_i(x,y),\ i = 1, 2, \dots, N$ represents an image of size $M \times M$. The population arises from the formation of the vectors $\underline{f}$ across the entire collection of pixels (x,y).

Consider now a population of vectors $\underline{g}$ of the form:

$$\underline{g}(x,y) = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ \vdots \\ g_N(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.

(i) Prove and explain the relationship between the covariance matrix of $\underline{g}(x,y)$ and the covariance matrix of $\underline{f}(x,y)$. What is the structure of the covariance matrix of $\underline{g}(x,y)$? **[3]**

(ii) Can the covariance matrix of $\underline{g}(x,y)$ be non-symmetric? Justify your answer. **[2]**

(iii) Suppose that $N = 8$ and that the eigenvalues of the covariance matrix of $\underline{f}(x,y)$ are:

$$[168 \quad 6.1 \quad 0.08 \quad 13 \quad 64 \quad 214 \quad 1.2 \quad 0.2]$$

What will be the mean square error if we use principal component images associated with the largest eigenvalues for 8: 1 and 4: 1 data compression? **[4]**

(c) Consider again Question 1(b) above in the case of 3 images ($N = 3$). The covariance matrix of the population is $\underline{C_f}$ with elements defined as $C_{m,n} = 0.5^{|m-n|}, 1 \le m,n \le 3$.

(i) Write explicitly the covariance matrix $\underline{C_f}$ of the population of vectors $\underline{f}$. **[3]**

(ii) What will be the mean square error if we use the principal component image from the set of images $g_1(x,y)$, $g_2(x,y)$ and $g_3(x,y)$ to reconstruct the original images? The eigenvalues of matrix $\underline{C_f}$ are $\frac{1}{8}\left(9+\sqrt{33}\right), \frac{1}{8}\left(9-\sqrt{33}\right), \frac{3}{4}$. **[3]**

2. (a) The histogram of a 3 bits/pixel square image is given in **Table 2.1** below.

Grey level	Number of pixels
0	4
1	4
2	0
3	0
4	0
5	3
6	2
7	3

Table 2.1

(i) Find out the size of the image and the total number of pixels. **[1]**

(ii) What would be the new histogram if we make the MSB (Most Significant Bit) of all pixels equal to 1? **[3]**

(iii) Create a binary image with pixel values 0 and 7 using the technique of **Thresholding**. Compute the threshold T such that $T = m$ where m is the mean grey level value of the original image with the histogram given in **Table 2.1** above. Apply this threshold and draw the new histogram for the output binary image. **[3]**

(b) Which statistical parameters indicate the brightness of a grey level image among Standard Deviation, Mean and Median? Justify your answer by examining separately each of the given parameters. **[5]**

(c) If the pixels of an image are rearranged which of the function/parameters below are affected? (i) Histogram, (ii) Mean, (iii) Median, (iv) Mode (v) None of the later. Justify your answer by examining separately each of the given parameters. **[4]**
[In statistics, the mode refers to the value that appears most frequently in a data set.]

(d) What should the sum of all elements in an edge detection mask be equal to? Justify your answer. **[4]**

3. We are given the degraded version $g(x, y)$ of a discrete image $f(x, y)$ such that in lexicographic ordering

$$g = Hf + n \quad (\mathbf{3.1})$$

where H is the degradation matrix which is assumed to be block-circulant and n is the noise term which is assumed to be zero-mean, white and independent of the image f. The symbols f, n, g denote the lexicographically ordered original, noise and degraded image respectively. All images involved have size $N \times N$ after extension and zero-padding.

(a) (i) Provide the expression of the Inverse Filter in frequency domain that we may use to restore the original image. Under what assumptions can we perfectly restore the image using the technique of Inverse Filtering? **[3]**

(ii) Explain why there might exist frequency pair(s) (u, v) for which Inverse Filtering cannot be applied. **[3]**

(iii) A particular image $f(x, y)$ is distorted by convolution with the space invariant point function $h(x, y)$ where:

$h(x, y) = \frac{1}{5}[\delta(x, y) + \delta(x-1, y) + \delta(x+1, y) + \delta(x, y-1) + \delta(x, y+1)]$

Show that for a specific value of N there may exist at least one discrete frequency pair (u, v) for which Inverse Filtering cannot be applied. Consequently, demonstrate how the method of Pseudoinverse Filtering for image restoration can be applied in this scenario. **[4]**

(b) The Constrained Least Squares (CLS) class of filters is designed to address potential issues that may arise during the application of Inverse Filtering to restore image f from **Equation (3.1)** above. Their mathematical formulation is depicted below:

$$\frac{H(u, v)^*}{|H(u, v)|^2 + \alpha |C(u, v)|^2} \quad (\mathbf{3.2})$$

where α is the regularizations parameter and $C(u, v)$ represents the Laplacian operator. Show that there aren't any frequency pairs for which the original image cannot be recovered using **Equation (3.2)** above. **[4]**

(c) Wiener Filtering is an alternative stochastic method used in image restoration to recover the image f from **Equation (3.1)** above.

(i) Describe any type of statistical information regarding the original image and the additive noise that is required for implementing Wiener Filtering in image restoration. **[3]**

(ii) Discuss potential strategies for obtaining the statistical information provided in (i) if it's not directly provided. **[3]**

4. (a) A binary image is to be Huffman-coded in blocks of M pixels. The successive pixel values are independent from each other. We assume that the pixel values are normalized, meaning that the maximum pixel value is 1. We know that 5% of the pixel values are 1 and the rest of the pixel values are 0. What are the Huffman codes for:

(i) $M = 1$ **[1]**

(ii) $M = 2$ **[2]**

(iii) $M = 3$ **[3]**

(iv) For each of the above three cases, calculate the coding efficiency and the redundancy of each codebook. **[2 points for each *M*; total 6 points]**

(b) A set of images is to be compressed. Each pixel of each image has a value in the range $[0, 3]$, i.e., we require 2 bits/pixel. An image taken from this set is given below. The frequency of occurrence of pixels of different values in this image, is typical of the set of images as a whole.

$$\begin{bmatrix} 3 & 3 & 3 & 2 \\ 2 & 3 & 3 & 3 \\ 3 & 2 & 2 & 2 \\ 2 & 1 & 1 & 0 \end{bmatrix}$$

What is the degree of compression achievable using the following methods?

(i) Huffman coding of the pixel values. **[4]**

(ii) Forming differences between adjacent pixels (assuming horizontal row-by-row scan) and then Huffman coding these differences. For the top-left pixel use the value of 0 in this calculation.

[4]